

Midterm Exam

Form A

Franklin University Switzerland · Instructor: Assoc. Prof. C. Burke · Friday 9 October 2026 · in class

Name: _____

How this exam works

- **The blueprint.** The exam follows the steps of our analysis blueprint: hypothesis → data → describe → relate and predict → compute in Sheets → interpret.
- **Time and points.** 50 points, about 45 minutes. The suggested time is printed in each part’s banner.
- **Show your work.** Partial credit is given for correct steps, even if a final answer is wrong. Round to 1 decimal place unless told otherwise.
- **Part A comes first.** Do it before you look at the data page.
- **What is marked.** In Parts A, E and F you are marked on clear, testable thinking and an honest reading of the evidence, not on whether your hypothesis turns out to be right.
- **Allowed:** pencil or pen, eraser, ruler, calculator (a phone may be used as a calculator only). **Not allowed:** notes, books, any other use of a phone, talking.

Part	What it covers	Points	Score
A	Hypothesis & prediction (before the data)	6	
B	Describe the data	11	
C	Relate, predict and check the errors	12	
D	Spreadsheets and LINEST	12	
E	Interpret & communicate	5	
F	Your own idea	4	
Total		50	

About the data

A seaside café recorded sales and weather for **168 consecutive trading days** (7 April–21 September 2025). These are **synthetic teaching data**, simulated for this course, not real café records. Your exam uses a sample of **10 of those days**; Parts D and E also use results from all 168 days. Each row is one day.

Variable	Concept: what we want to know	Operationalization: what was measured
Temperature (X)	How warm (sunny) the day was	Daily mean outdoor temperature in °C during opening hours
Gelato units (Y)	How much gelato customers buy	Number of standard servings of gelato sold that day (no stock-outs or closures)
weekend	Weekend trading	1 if Saturday or Sunday, 0 otherwise
beach_event	A crowd-pulling event nearby	1 if an organised public event took place on the nearby beach, 0 otherwise

PART A · Hypothesis & prediction (before the data)

≈ 5 min · 6 pts

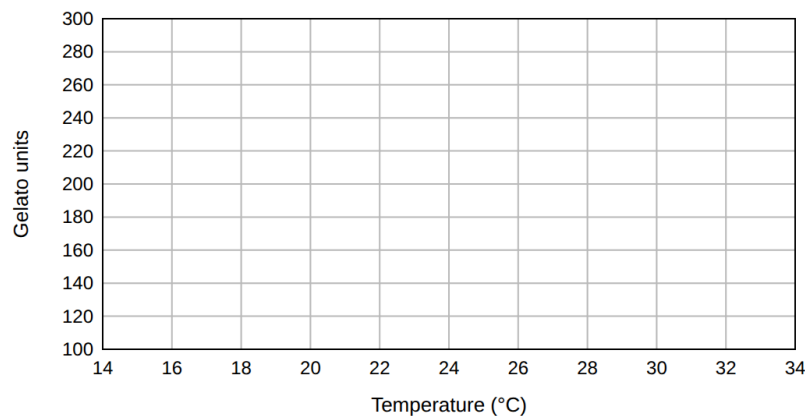
Research question. Across trading days, is warmer weather related to how much gelato the café sells? (Temperature = X, the explanatory variable; Gelato units = Y, the outcome.)

- A1** Write a **testable hypothesis** about temperature_c and daily gelato units. Your hypothesis must (i) say which **direction** the relationship goes, and (ii) name the **unit of analysis** (one trading day). Then (iii) describe a **result** in the data that would **falsify** it.

2 pts

- A2** **Predict before you look.** On the empty axes, draw the pattern you **expect** if your hypothesis is true: about 8 dots and your expected trend line. Then name **one other variable** from the table above that could also push gelato sales up or down on a day, and say which way.

2 pts



- A3** **Data quality.** (a) Customers “wanting gelato” is the concept; **units sold** is the measure. Give one way the measure could be **invalid** for the concept. (b) In July the weather sensor is moved from a shaded wall to a sunny one. Is this mainly a **validity** or a **reliability / comparability** problem? Explain in one sentence.

2 pts

Stop. Finish Part A before you turn the page.

PART B · Describe the data

≈ 9 min · 11 pts

The table lists a sample of 10 days in date order. In this part you describe **Gelato units** (Y). The last two columns are for B2.

#	Date	Temp (°C)	Weekend	Event	Gelato units	Units – mean	(Units – mean) ²
1	Apr 8	22.2	0	0	168		
2	Apr 18	22.3	0	0	181		
3	May 10	21	1	0	227		
4	May 15	23.5	0	0	191		
5	May 19	27.4	0	0	199		
6	May 21	24	0	0	186		
7	May 26	23.4	0	0	165		
8	Aug 2	30.9	1	1	294		
9	Aug 30	21.4	1	0	203		
10	Sep 14	19.9	1	1	176		

Reference (use on any question)

mean = $\Sigma \text{values} \div n$ · sample sd: $s = \sqrt{[\Sigma(\text{value} - \text{mean})^2 \div (n - 1)]}$ · $z = (\text{value} - \text{mean}) \div s$ · flag a value as an outlier if $|z| > 2$

slope $b = S_{xy} \div S_{xx}$ · intercept $a = \bar{y} - b \cdot \bar{x}$ · prediction $\hat{y} = a + b \cdot x$ · $r = S_{xy} \div \sqrt{S_{xx} \cdot S_{yy}}$ · $R^2 = r^2$ ($S_{xx} = \Sigma(x - \bar{x})^2$, $S_{xy} = \Sigma(x - \bar{x})(y - \bar{y})$, $S_{yy} = \Sigma(y - \bar{y})^2$)

residual = actual – predicted · mean absolute error (MAE) = average of $|\text{residual}|$ · weighted mean = $\Sigma(\text{weight} \times \text{value}) \div \Sigma \text{weights}$

difference of means: $SE = \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$, $t = (\text{mean}_2 - \text{mean}_1) \div SE$ · treat $|t| > 2$ as statistically significant (5% level)

coefficient $t = \text{coefficient} \div \text{its standard error}$ · LINEST(y, X, TRUE, TRUE) returns, for X columns in order 1...k: Row 1

coefficients **in reverse order** (column k first ... column 1, intercept last) · Row 2 their standard errors · Row 3: R^2 , SE of y (typical size of a residual) · Row 4: F, df · Row 5: SS regression, SS residual

risk = likelihood × consequence

B1 Centre. Calculate the **mean** and the **median** of Gelato units. $\Sigma = \underline{\hspace{2cm}} \div 10$

2 pts

Mean	
Median	

B2 Spread. Use your mean from B1 to complete the last two columns of the table. Then fill in the results below (1 decimal place).

3 pts

$\Sigma (\text{Units} - \text{mean})^2$ (sum of your last column)	
Variance = $\Sigma \div (n - 1)$	
$s = \sqrt{\text{variance}}$	

B3 Conditional means (what AVERAGEIFS does). Using the Weekend column, find the number of days and the **mean Gelato units** for **weekend days** (Weekend = 1) and for **weekdays** (Weekend = 0).

2 pts

Group	Days (n)	Mean Gelato units
Weekend (1)		
Weekday (0)		

B4 Weighted average. Over the whole season there were **120 weekdays** and **48 weekend days**. Use your two group means from B3 to calculate the season-wide **weighted mean** Gelato units per day. In one sentence, say why this is better than the plain mean of your 10 days.

2 pts

B5 Outlier check. Calculate the z-score of the day with the **most extreme** Gelato units (use your B1 and B2 answers). Using the rule $|z| > 2$, is it an outlier? If it is a flagged day, would you **remove it, keep it, or report results both ways**? Justify in one sentence.

2 pts

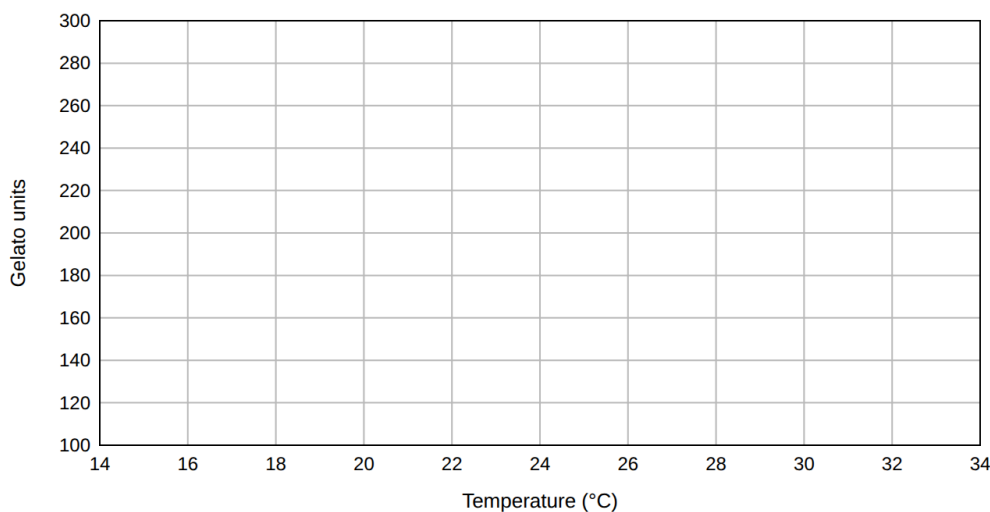
PART C · Relate, predict and check the errors

≈ 10 min · 12 pts

Use the grid below: Temperature on the horizontal axis and Gelato units on the vertical axis. Sheets reports these summary values for the 10 days: $\bar{x} = 23.6$, $S_{xx} = 96.88$, $S_{xy} = 766.20$, $S_{yy} = 13028$; $\bar{y}$ is your B1 answer.

- C1** Plot all 10 days as dots. Mark the **point of means** ($\bar{x}$, $\bar{y}$) with a $\times$. Then draw the **line of best fit by eye**: it should pass close to the $\times$, with about as many dots above the line as below.

3 pts



- C2 Regression by hand.** Calculate the slope $b = S_{xy} \div S_{xx}$ (2 decimals) and the intercept $a = \bar{y} - b \cdot \bar{x}$. Write the equation $\hat{y} = a + b \cdot x$, then complete: “One more °C goes with about _____ servings gelato per day.”

3 pts

- C3 Correlation.** Calculate $r = S_{xy} \div \sqrt{(S_{xx} \cdot S_{yy})}$ and R^2 . Say in plain words what R^2 tells you here (not “the line is right R^2 of the time”).

2 pts

- C4 Predictions and residuals.** Use your line from C2 (round a and b to 2 decimals). For the four days below, complete the table: predicted Gelato units, **residual** (actual – predicted) and absolute residual. Then calculate the **MAE** of the four predictions and name the day the model predicts **worst**. 4 pts

Day	Temp	Actual	Predicted $\hat{y}$	Residual	Residual
Apr 8	22.2	168			
May 10	21	227			
May 15	23.5	191			
May 21	24	186			

MAE of the four predictions	
Worst-predicted day	

PART D · Spreadsheets and LINEST

≈ 10 min · 12 pts

The 10 days are in a Google Sheet: row 1 holds headers and rows 2–11 hold the days. Columns: **A** date, **B** temperature_c, **C** weekend, **D** beach_event, **E** gelato_units. Part of the sheet is shown.

	A	B	C	D	E	F	G	H
1	date	temperature_c	weekend	beach_event	gelato_units		Statistic	Value
2	Apr 8	22.2	0	0	168		Mean units, weekend days	
3	Apr 18	22.3	0	0	181		Weekend event days (count)	
4	May 10	21	1	0	227		Predicted at 26	
5	May 15	23.5	0	0	191		Correlation	
6	May 19	27.4	0	0	199			
7	May 21	24	0	0	186			
8	May 26	23.4	0	0	165			
9	Aug 2	30.9	1	1	294			
10	Aug 30	21.4	1	0	203			
11	Sep 14	19.9	1	1	176			

- D1 Write the formula to type** in each cell. Any correct form is accepted. 3 pts

Cell	It should calculate...	Formula to type
H2	the mean of column E on days where weekend = 1	
H3	the number of days where weekend = 1 and beach_event = 1	
H4	the predicted value of E for a day with B = 26, using the 10 days	

- D2** Cell H5 should show the correlation between column B and column E, but a student typed **=CORREL(B2:B10, E2:E11)** and Sheets shows an error. (a) What is wrong? (b) Write the corrected formula. 2 pts

LINEST output. Using **all 168 days**, a student fitted a multiple regression of **gelato_units** on **temperature_c**, **beach_event**, **weekend** (X columns in that order) with **=LINEST(y, X, TRUE, TRUE)**. Sheets returned:

	Col 1	Col 2	Col 3	Col 4
Row 1	34.16	25.74	5.57	48.97
Row 2	3.85	4.27	0.52	11.86
Row 3	0.61	22.52	#N/A	#N/A
Row 4	86.39	164.00	#N/A	#N/A
Row 5	131402.67	83146.74	#N/A	#N/A

- D3 Reading the coefficients.** (a) Which column of Row 1 holds the **temperature_c** coefficient, and what is its value? (b) Interpret it in one sentence, **holding the other variables constant**. (c) What does the number in Row 1, Col 4 represent? 3 pts

- D4 Evidence and error.** (a) Calculate the **t-statistic** for **beach_event** and say whether it is statistically significant ($|t| > 2$). (b) The standard error of y in Row 3 is a typical size of a residual. Say what it means for a day's gelato forecast. 2 pts

- D5 Predict and check.** On Jul 25: **temperature_c** = 28.3, **beach_event** = 1, **weekend** = 0; the café sold **243** units of gelato. Use the LINEST coefficients to calculate the **predicted** units and the **residual**. Is the residual larger than the typical size (Row 3, Col 2)? What does that tell you about this day? 2 pts

PART E · Interpret & communicate

≈ 5 min · 5 pts

- E1** On 30 June the café raised the gelato price from €4.00 to €4.50. Daily gelato units over all 168 days: **before** (n = 85) mean 183.2, sd 36.3; **after** (n = 83) mean 199.9, sd 33.5. (a) Calculate SE and t for the difference in means; is it statistically significant? (b) Give one reason the t-test alone cannot tell us the **price change caused** the difference, and name one variable you would add to a LINEST model to help. 3 pts

- E2 Uncertainty and risk.** Tomorrow is a hot (30 °C), event-day weekend (**beach_event** = 1, **weekend** = 1). (a) Use the LINEST model to give the **predicted** units and a range of **prediction ± 2 × SE** (SE = Row 3, Col 2). (b) Running out costs €4 profit per lost sale; unsold stock wastes €1.20 per serving. Which end of the range would you prepare for, and why? (risk = likelihood × consequence) 2 pts

PART F · Your own idea

≈ 4 min · 4 pts

A new variable might help explain gelato sales: **rainfall_mm**, **wind_kmh**, **humidity_pct**, **sunshine_hours** or the **posted price**. Choose **one** and write a mini-blueprint: (i) your **hypothesis** and why; (ii) the **expected sign** of its coefficient if added to the LINEST model above; (iii) which numbers in the LINEST output you would check (coefficient, SE, t, R^2 , SE of y, residuals) and what you would look for; (iv) a result that would **falsify** you. Marked on clear, testable reasoning, not on being right.

F1 Your mini-blueprint:

4 pts
